USN. [18MTSE201]

SECOND SEMESTER M.TECH EXAMINATION, JUNE–2019

ADVANCED CONCRETE TECHNOLOGY

Time:3 Hours **Maximum Marks:100**

Instructions:
- **i. Answer FIVE full questions, Question 1 and 2 are compulsory.**
- **ii. Missing data, if any, may be suitably assumed.**
- **iii. Use of IS 10262 & IS 456 is permitted**
- **iv. Use of DOE chart, erntroy and shacklock's permitted**

1. a. Discuss the phenomena of hydration of cement with their compound reaction and products. **10**

b. Enlist different types of cement. Write properties and application of following cement type; **10**

- **i.** Sulphate resisting cement
- **ii.** Low heat Portland cement
- **iii.** High alumina cement

2. a. Compare the relative merits and demerits of any 5 workability tests. **10**

b. Describe the most important factors that affect shrinkage and creep. **10**

3. a. Enlist the requirements of concrete performance generally demanded by clients **08**

b. Explain briefly any top three applications of high performance concrete (HPC). **12**

OR

4. Design a High Strength Concrete (HSC) by Erntroy shacklock method for use in the production of precast prestressed concrete to suit the following requirements: **20**

Specified 28-day works cube strength = 50 MPa

Very good degree of control; control factor = 0.80

Degree of workability = very low

Type of cement = ordinary Portland cement

Type of coarse aggregate = crushed granite (angular) of maximum size 10mm.

Type of fine aggregate = natural sand

Specific gravity of sand = 2.60

Specific gravity of cement = 3.15

Specific gravity of coarse aggregates = 2.50

Fine and coarse aggregates contain 5 and 1 percent moisture respectively and have grading characteristics as follows [Refer Figure:1, 2 and 3 and Table.1]

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IS sieve size		20 mm	10 mm	4.75 mm	2.36 mm	1.18 mm	600 µm	300 µm	150 µm
% Passing	Coarse aggregate	100	96	8	-	-	-	-	-
	Fine aggregate	-	100	98	80	65	50	10	0

5. a. What do you meant by self-compacting concrete (SCC)? Describe the characteristics of self-compacting concrete in the fresh condition. How is the flowability of SCC measured? **12**

   b. Enlist the polymers used in concrete composites. Describe polymer concrete and its applications. **08**

OR

6. a. Design a concrete mix by DOE method for a reinforced concrete work which will be exposed to the moderate condition and slump of 75mm is assumed. The concrete is to be designed for a mean compressive strength of 30MPa at the age of 28 days. A requirement of 25 mm cover is prescribed. Maximum size of aggregate is 20 mm uncrushed aggregate will be used. Sieve analysis shows that 50% passes through 600µ sieve. The bulk specific gravity of aggregate is found to be 2.65 [Refer Figure 4, 5 & 6 and Table 2, 3 & 4] **20**

7. a. What are the basic objectives and assumptions of formwork design and what are the aspects that should be considered to achieve these objectives? **10**

   b. Enlist the various systems of formwork and Describe heavy duty tower system and stair tower system. **10**

OR

8. a. List some of the common deficiencies in formwork which lead to the failure of structures. What precautions should be taken to prevent failure? **10**

   b. List the different types of shores and explain merits and demerits of wood post shore and adjustable shores. **10**

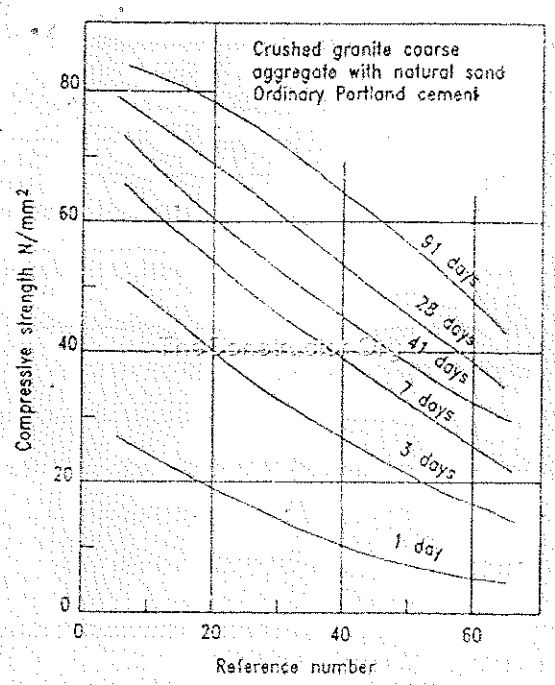

Fig-1: Relation between compressive strength And reference number (Erntroy and Shacklock)

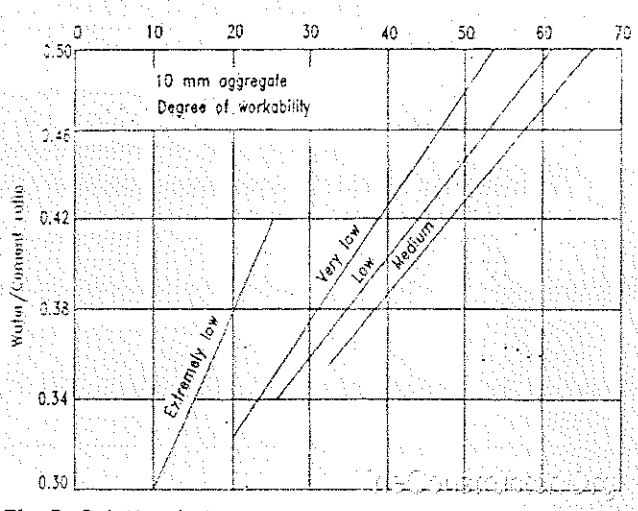

Fig-2: Relation between water-cement ratio and Reference No.

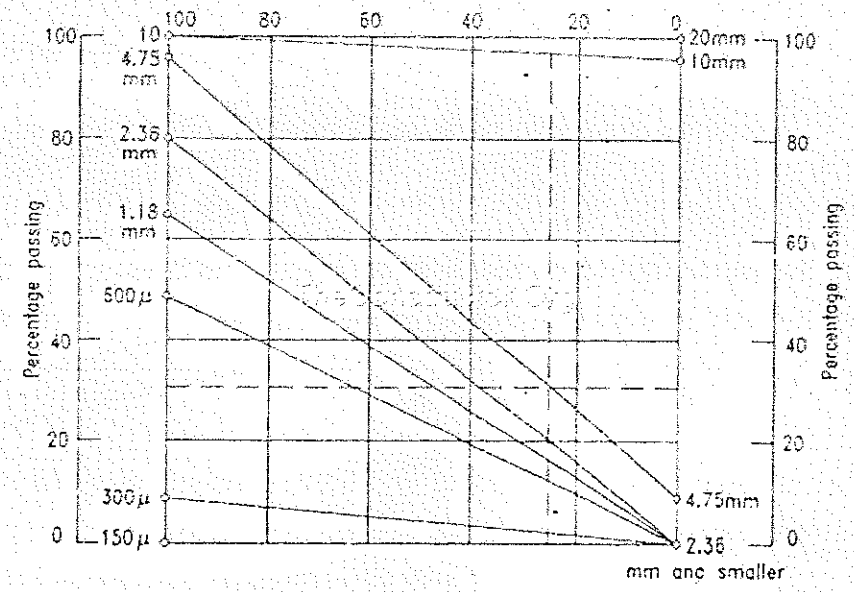

Fig-3: Combining of Fine aggregates and Coarse aggregates

Table.1: Aggregate cement ratio (by weight) required to give four degrees of workability with different water cement ratios using ordinary Portland cement

*Type of coarse aggregate**	*Irregular gravel*								*Crushed granite*							
Maximum size of aggregate	*20 mm*				*10 mm*				*20 mm*				*10mm*			
*Degree of workability***	*EL*	*VL*	*L*	*M*	*EL*	*VL*	*L*	*M*	*EL*	*VL*	*L*	*M*	*EL*	*VL*	*L*	*M*
Water/cement ratio by weight																
0.30	3.0	-	-	-	2.1	-	-		3.3	-		-	2.9	-	-	
0.32	3.8	2.5	-	-	3.2		-		4.0	2.6		-	3.6	2.9		
0.34	4.5	3.0	2.5	-	3.9	2.6	-		4.6	3.2	2.6	-	4.2	3.8	2.5	
0.36	5.2	3.5	3.0	2.5	4.6	3.1	2.6		5.2	3.6	3.1	2.6	4.7	3.2	2.7	2.3
0.38	-	4.0	4.1	2.9	5.2	3.5	3.0	2.5	-	4.1	3.5	2.9	5.2	3.6	3.0	2.6
0.40	-	4.4	3.8	3.2	-	3.9	3.3	2.7	-	4.5	3.8	3.2	-	4.0	3.3	2.9
0.42	-	4.9	4.1	3.5	-	4.3	3.6	3.0	-	4.9	4.2	3.5	-	4.4	3.6	3.1
0.44	-	5.3	4.5	3.8	-	4.7	3.9	3.5	-	5.3	4.5	3.7		4.8	3.9	3.3
0.46	-	-	4.8	4.0	-	5.1	4.2	3.6	-	-	4.8	4.0	-	5.1	4.2	3.6
0.48	-	-	5.2	4.4	-	5.4	4.5	3.8	-	-	5.1	4.2		5.5	4.5	3.8
0.50	-	-	5.5	4.7	-	-	4.8	4.0	-	-	5.4	4.5	-		4.7	4.0

* Natural sand used in combination with both types of coarse aggregate

** EL = extremely low
VL = very low
L = low
M = medium

Table.2: Approximate Compressive Strength of Concrete Made with a free Water/Cement Ratio of 0.50. According to the 1988 British Method

Type of Cement	Type of C.A	Compressive Strength at the age (cube) of days MPa			
		3	7	28	91
Ordinary Portland cement (Type I)	uncrushed	22	30	42	49
Sulphate Resisting Cement (Type V)	Crushed	27	36	49	55
Rapid-Hardening Portland Cement (Type III)	Uncrushed	29	37	48	54
	Crushed	34	43	55	61

Table.3: Approximate Free Water Contents Required to Give Various Levels of Workability by British method

Aggregate		Water Content kg/m³ for:			
Max-size mm	Type	Slump 0-10 / Vebe > 12 seconds	10-30 / 6-12	30-60 / 3-6	60-180 / 0-3
10	Uncrushed	150	180	205	225
	Crushed	180	205	230	250
20	Uncrushed	135	160	190	195
	crushed	170	190	210	225
40	Uncrushed	115	140	160	175
	crushed	155	175	190	205

Fig.4: Relation between compressive strength and free water/cement ratio for use in British method

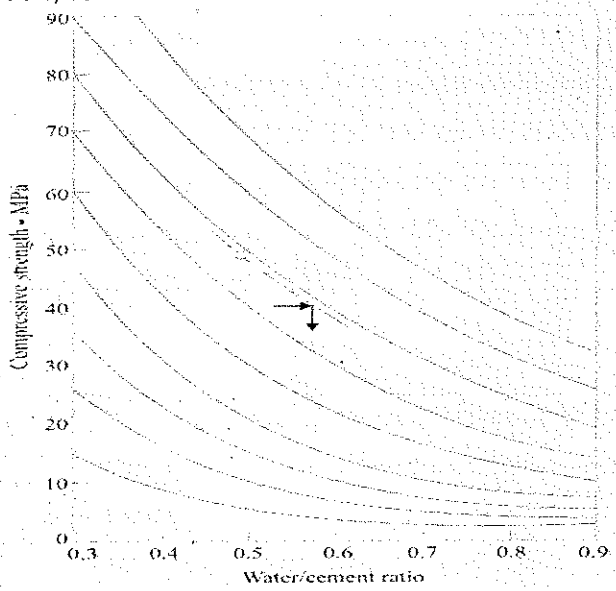

Fig.5: Estimated wet density for fully compacted concrete

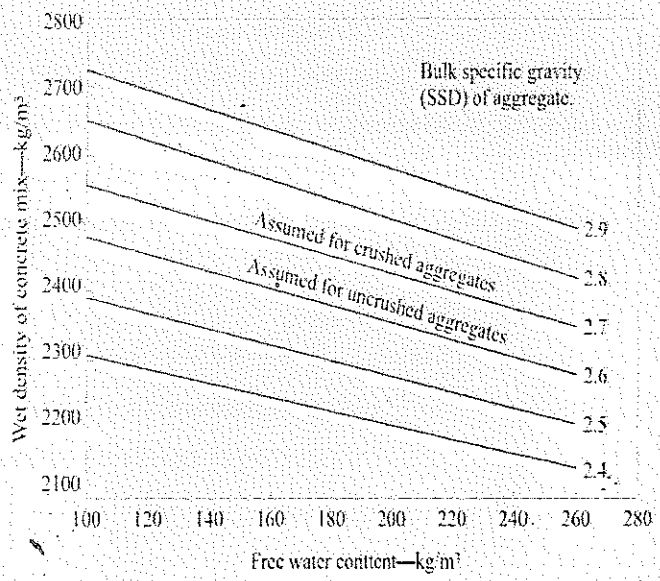

Table.4: Requirements of BS 8110: Part I: 1985 to Ensure Durability under Specified Exposure Conditions of Reinforced and Prestressed Concrete Made with Normal Weight Aggregate

Condition of exposure	Nominal Cover of Concrete in mm				
Mild	25	20	20	20	20
Moderate	–	35	30	25	20
Severe	–	–	40	30	25
Very Severe	–	–	50	40	30
Extreme	–	–	–	60	50
Maximum Water/ Cementitious material ratio	0.65	0.60	0.55	0.50	0.45
Minimum content of cementitious Material in kg/m³	275	300	325	350	400
Minimum grade MPa	30	35	40	45	50
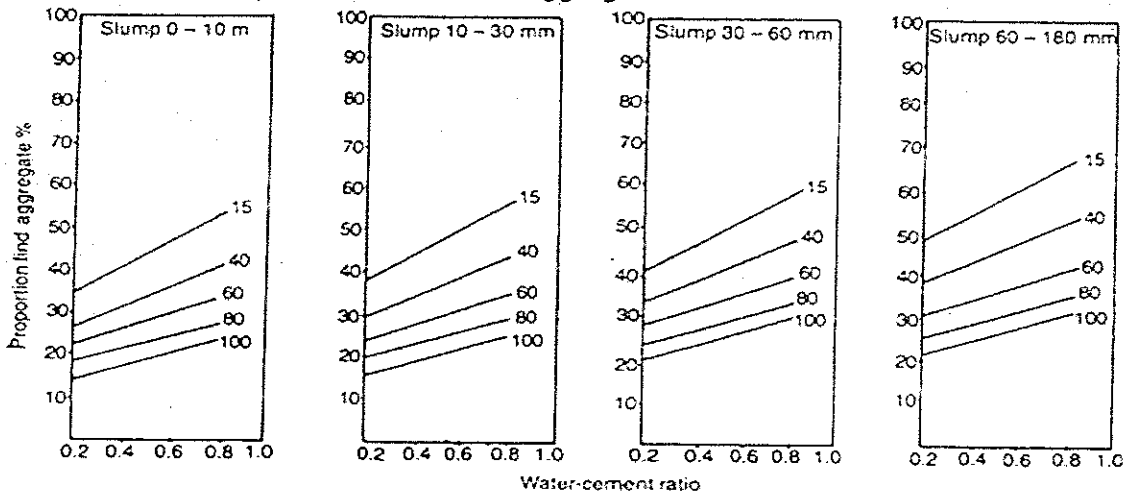
Fig.6: Percentage of fine aggregate in total aggregate as a function of free water/cement ratio for various values of workability and 20mm size of aggregate.